

# Week 6 – Machine Learning Model Architecture Design (20260611)

## Week 6 – Machine Learning Model Architecture Design

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> This is the active Week 6 report for the current run. Older dated Week 6 > report files in `data/reports/` are treated as superseded outputs.

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### 1. Executive Summary

This document describes the machine learning architecture designed and implemented in Week 6 of the Quantitative Research & Trading project. Two complementary approaches price chooser options on JPM stock:

- **Approach 1 (ML + chooser closed form):** ML models predict 20-day forward realised volatility; the predicted  $\sigma$  is fed into the chooser pricing formula.
- **Approach 2 (End-to-End):** ML models directly map chooser contract parameters and market features to chooser prices.

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### 2. Problem Formulation

#### Approach 1

$$\hat{V}_{\text{chooser}} = C \left( S, K, T_2, r, q, \hat{\sigma}_{\text{ML}} \right) + P \left( S, K e^{-r(T_2 - T_1)}, T_2 - T_1, r, q, \hat{\sigma}_{\text{ML}} \right)$$

where  $\hat{\sigma}_{\text{ML}}$  is the ML-predicted 20-day forward realised volatility.

$$\text{Target: } \sigma_{\text{fwd}, 20d}[t] = \sqrt{252} \cdot \text{std} \left( \ln \frac{S_{t+i}}{S_{t+i-1}} \right)_{i=1}^{20}$$

#### Approach 2

$$\hat{V}_{\text{chooser}} = f_{\theta} \left( S, K, T_1, T_2, r, q, m, \mathbf{x}_{\text{market}} \right)$$

where  $f_{\theta}$  is a trained ML model and  $\mathbf{x}_{\text{market}}$  is the market feature vector.

**Target:** chooser price benchmarked with Monte Carlo-valued call/put legs.

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### 3. Data & Feature Engineering

### 3.1 Raw Data Sources

| Source                              | Description             | Frequency   |
|-------------------------------------|-------------------------|-------------|
| yahoo_jpm_2018_2024.csv             | JPM daily close prices  | Daily       |
| fred_DGS10_2018_2024.csv            | US 10-yr Treasury yield | Daily       |
| fred_VIXCLS_2018_2024.csv           | CBOE VIX index          | Daily       |
| jpm_dividends_2018_2024.csv         | JPM dividend payments   | Per event   |
| alphavantage_news_jpm_2018_2024.csv | News sentiment scores   | Per article |

### 3.2 Market Features (Approach 1 – Volatility Dataset)

All features are strictly **\*\*backward-looking\*\*** to prevent look-ahead bias. Dataset: 1,681 trading days.

- `hist\_vol\_5d`
- `hist\_vol\_10d`
- `hist\_vol\_20d`
- `hist\_vol\_40d`
- `hist\_vol\_60d`
- `vol\_ratio\_5\_20`
- `vol\_ratio\_20\_60`
- `vol\_20d\_change`
- `return\_1d`
- `return\_5d`
- `return\_20d`
- `price\_to\_ma\_20d`
- `price\_to\_ma\_60d`
- `vix`
- `vix\_change\_5d`
- `vix\_ma\_ratio`
- `vix\_jpm\_corr\_20d`
- `r`
- `q`
- `sentiment\_7d`
- `sentiment\_20d`
- `news\_count\_7d`
- `drawdown\_20d`

### 3.3 Features (Approach 2 – Chooser Pricing Dataset)

All market features above, plus chooser-specific parameters. Dataset: 17,850+ rows (daily dates × chooser contracts).

- `vol\_ratio\_5\_20`

- `vol\_ratio\_20\_60`
- `vol\_20d\_change`
- `return\_1d`
- `return\_5d`
- `return\_20d`
- `price\_to\_ma\_20d`
- `price\_to\_ma\_60d`
- `vix`
- `vix\_change\_5d`
- `vix\_ma\_ratio`
- `vix\_jpm\_corr\_20d`
- `r`
- `q`
- `sentiment\_7d`
- `sentiment\_20d`
- `news\_count\_7d`
- `drawdown\_20d`
- `S`
- `K`
- `moneyness`
- `T1`
- `T2`

### 3.4 Target Variables

| Approach | Target        | Description                                       |
|----------|---------------|---------------------------------------------------|
| 1        | fwd_vol_20d   | Annualised realised vol over next 20 trading days |
| 2        | chooser_price | Chooser price using Monte Carlo benchmark         |

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## 4. Time-Series Validation Framework

Data is split **chronologically** (never randomly) to prevent look-ahead bias. All features are scaled using `RobustScaler` fitted **only** on the training set. For model selection, each tunable learner is optimized with `RandomizedSearchCV` and `TimeSeriesSplit`, then refit on the combined train+validation partition before the final test-set evaluation.

| Split      | Date Range              | Fraction |
|------------|-------------------------|----------|
| Train      | 2018-03-29 → 2022-11-28 | 70%      |
| Validation | 2022-11-29 → 2023-11-29 | 15%      |
| Test       | 2023-11-30 → 2024-12-02 | 15%      |

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## 5. Model Architectures

### 5.1 Approach 1 – ML Volatility Prediction

| Model        | CV MAE  | Val MAE | Best parameters                                                                                                                                                |
|--------------|---------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RandomForest | 0.12686 | 0.07558 | n_estimators=200,<br>min_samples_split=10,<br>min_samples_leaf=6,<br>max_features=sqrt,<br>max_depth=8                                                         |
| XGBoost      | 0.14393 | 0.07101 | subsample=0.8,<br>reg_lambda=1.0,<br>reg_alpha=1.0,<br>n_estimators=200,<br>min_child_weight=7,<br>max_depth=3,<br>learning_rate=0.01,<br>colsample_bytree=0.9 |
| LSTM         | nan     | 0.04836 | -                                                                                                                                                              |

#### LSTM

- Architecture: LSTM(64) → Dropout(0.2) → LSTM(32) → Dropout(0.2) → Dense(16) → Dense(1)
- Lookback window: 20 trading days
- Optimizer: Adam | Loss: MSE | Early stopping (patience=10)
- Input shape: (batch, 20, 23)

### 5.2 Approach 2 – End-to-End Supervised Chooser Pricing

| Model            | CV MAE  | Val MAE | Best parameters                                                                                                                                               |
|------------------|---------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LinearRegression | 7.81781 | 4.74567 | model__positive=True,<br>model__fit_intercept=True                                                                                                            |
| XGBoost          | 9.71810 | 5.83563 | subsample=0.9, reg_lambda=5.0,<br>reg_alpha=0.01,<br>n_estimators=1200,<br>min_child_weight=1,<br>max_depth=6,<br>learning_rate=0.03,<br>colsample_bytree=0.8 |
| NeuralNetwork    | 5.78848 | 6.07188 | model__learning_rate_init=0.001,<br>model__hidden_layer_sizes=(256,<br>128, 64, 32),<br>model__batch_size=64,<br>model__alpha=0.001                           |

#### Baseline comparison

- Week 4 BSM baseline (European options): MAE=0.112135, RMSE=0.174916, p-value=0.9524
- The closed-form chooser formula computed with maturity-matched historical volatility is used only as a comparison baseline, not as the training target.

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## 6. Performance Summary (Test Set)

### 6.1 Approach 1 – Volatility Prediction

| Model        | Vol MAE | Vol RMSE | Option MAE | Option RMSE |
|--------------|---------|----------|------------|-------------|
| RandomForest | 0.07018 | 0.09641  | 7.1302     | 9.6827      |
| XGBoost      | 0.06879 | 0.09608  | 7.9650     | 11.0598     |
| LSTM         | 0.07472 | 0.10097  | 8.0513     | 11.3817     |

\*Vol MAE/RMSE: annualised vol units. Chooser MAE/RMSE: USD.\*

### 6.2 Approach 2 – End-to-End Chooser Pricing

| Model            | MAE (\$) | RMSE (\$) | R <sup>2</sup> |
|------------------|----------|-----------|----------------|
| LinearRegression | 11.2689  | 14.2303   | -0.2552        |
| XGBoost          | 12.5877  | 16.2446   | -0.6357        |
| NeuralNetwork    | 8.4856   | 11.1067   | 0.2354         |
| BSM Baseline     | 0.6531   | 0.9058    | 0.9949         |

\*Target: chooser price benchmarked with Monte Carlo-valued call/put legs. The table includes the closed-form BSM baseline for the same contracts.\*

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## 7. Limitations & Recommended Next Steps

### Current Limitations

#### #### Data & Targets

- **\*\*No market-implied vol\*\***: Targets are derived from historical/closed-form

prices, not actual market option quotes. This is a fundamental constraint: Approach 2 learns to predict a synthetic chooser surface, not a real market surface. High R<sup>2</sup> values (e.g., MLP R<sup>2</sup>=0.89) reflect model fit to the synthetic label, NOT market predictability.

- **\*\*Synthetic chooser labels (Approach 2)\*\***: Targets are computed as

``chooser_price(S, K, T1, T2, r, q, hist_vol_match(T2))``. Since this is a deterministic closed-form calculation, linear models and neural networks with sufficient capacity can still fit the surface, but the direct historical-vol feature leakage into the input set has been removed.

- **\*\*Volatility term-structure matching\*\***: Vol used for each maturity is now

matched to option T2, preventing term-structure mismatch. Approach 2 now includes volatility proxy features such as VIX, volatility ratios, and volatility change signals, while excluding raw historical volatility windows.

#### #### Model Architecture

- **\*\*Simplified chooser grid\*\***: 2 decision times × 3 maturities × 3 moneyness levels.
- **\*\*Static features\*\***: No real-time microstructure data (bid-ask, volume).

- **No market data feedback**: Models do not adjust based on actual prices observed at decision time  $T_1$ .

## Model-Specific Notes

### #### Approach 2 – Route 2 Pricing

- **LinearRegression & MLP**:  $R^2$  should improve materially once volatility proxy features are included, because these models now receive the strongest explanatory signal that is allowed by the experiment design.
- **XGBoost/GradientBoosting**: Lower  $R^2$  (0.40-0.50) may indicate that tree models struggle with the smooth closed-form surface; alternatively, the current feature set lacks sufficient flexibility for tree splits. This is NOT a sign of data leakage or feature engineering failure—it reflects model-surface mismatch.

## Recommended Next Steps (Week 6+)

- Incorporate implied volatility data for more realistic targets.
- Add Greeks (delta, gamma, vega) as engineered input features.
- Implement Bayesian hyperparameter optimisation.
- Extend LSTM to multi-step ahead vol forecasting.
- Build ensemble that combines Approach 1 and Approach 2 predictions.
- Evaluate on 2025+ data for out-of-sample performance.
- **Collect real market chooser prices** and retrain Approach 2 models on actual traded prices instead of synthetic targets.

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\*Generated by `week5\_ml\_models.py` | v1.0 | 20260611\*